

Bhanu Nagpure

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Android Developer — Kotlin · Jetpack Compose · On-Device AI · Open to On-site / Remote / Hybrid

Profile

Android engineer with **2+ years** of hands-on experience building and shipping production-grade apps on Google Play. Founder of **Devanshu Studios** — independently conceived and launched **2 apps**, including an **on-device AI chatbot** (LiteRT-LM) and a cross-platform learning platform. Deep expertise in **Kotlin, Jetpack Compose (Material 3), MVVM/MVI, Hilt, and custom camera pipelines (CameraX)**. International research intern (Mitacs Globalink, Canada). Published researcher, patent applicant, open-source contributor, and technical writer on Medium.

Core Technical Skills

Android: Kotlin, Java, Jetpack Compose (Material 3), Android SDK, MVVM/MVI, Hilt (DI), WorkManager, CameraX, Room, Proto DataStore, EncryptedSharedPreferences

On-Device AI: LiteRT-LM (TFLite runtime), Gemma 2/3, Phi 3.5 Mini, SmolLM, MediaPipe FaceMesh, Google ML Kit

Data & Async: Kotlin Coroutines & Flow, Retrofit, REST APIs, Proto DataStore, Room

Firebase & Monetisation: Firebase (Auth, Crashlytics, Analytics), RevenueCat (IAP/subscriptions), Google AdMob

Cross-Platform: Flutter, Dart, Riverpod 2.0 (secondary stack — shipped to production)

Languages: Kotlin, Java, Dart, Python, C++, SQL

Tools: Git, GitHub, Android Studio, Gradle (Kotlin DSL), CI/CD, Google Cloud Platform, Linux

Education

G H Raisoni College of Engineering

Bachelor of Technology in Computer Science & Engineering | CGPA: 8.7 / 10.0

Nagpur, India

Dec 2021 – Jun 2025

Professional Experience

Founder & Lead Android Engineer

Aug 2025 – Present

Devanshu Studios

Remote

- Conceived, built, and shipped **Chakuli — Offline AI Chatbot** (Android, closed beta on Play Store): a **privacy-first on-device AI assistant** running Gemma 2/3, Phi 3.5 Mini, and SmolLM entirely via **LiteRT-LM** — zero cloud calls for AI inference.
- Engineered the complete **Jetpack Compose** (Material 3) UI, **Hilt DI**, **Proto DataStore**, **WorkManager** background model downloads, custom **CameraX** pipeline, Agent Skills system, audio chat, and encrypted key storage.
- Independently architected and shipped **PyMaster** (Live on Play Store) — a **50k+ LOC Flutter** Python learning platform with Clean Architecture, **Riverpod 2.0**, Flutter Flavors, and a **Gemini AI Tutor**; full RevenueCat monetisation stack.
- Managed the entire Android app lifecycle: architecture, Gradle (Kotlin DSL) build config, CI/CD pipeline, Play Console publishing, and Firebase Crashlytics monitoring.

Software Developer Intern

Oct 2022 – May 2024

BETIC-GHRCE (Business & Entrepreneurship Technology Innovation Centre)

Nagpur, India

- Built a production **Android (Kotlin/Java)** app for TMJ disorder screening, integrating **MediaPipe FaceMesh API** for real-time facial landmark detection — improved diagnostic classification accuracy by **60%**.
- Optimised **Google ML Kit** camera pipeline, achieving a **30%** performance boost and enabling sub-second inference for clinical real-time use.
- Contributed to patent filing: “*Smart Chairside TMJ Examination and Muscle Activity Detector Tool*” (Pub. No. 202321022476, Filed Mar 2023 — **Granted**).

Open Source Contributor

Feb 2025 – Present

Wikimedia Foundation — Scribe-Android

Remote

- Developed tablet-specific keyboard layouts in **Kotlin** for the Scribe-Android open-source keyboard, covering 7+ languages (German, French, Spanish, and others).

- Resolved critical UI bugs related to theme consistency and animations; refactored the core **KeyHandler** module, improving maintainability and reducing coupling.

Globalink Research Intern

Mitacs

Apr 2024 – Aug 2024

Edmonton, AB, Canada

- Led end-to-end development of a **cross-platform (iOS, Android, Web)** field-trip app using Flutter and Firebase; presented at the **3rd Annual TRESL Lab International Symposium**, Canada (Jul 2024).

Machine Learning Intern

Athabasca University

Aug 2024 – Oct 2024

Athabasca, AB, Canada

- Delivered **95% prediction accuracy** for a text ideology identification model using **Transformers / BERT**; deployed a prototype web interface for researcher validation.

Fellowship — Design for Health

IIT Bombay

Mar 2022 – Oct 2022

Mumbai, India

- Led a multidisciplinary team to prototype a medical device for xerostomia, reducing diagnostic time by **80%**. Built using MIT App Inventor and IoT hardware; presented at IIT Bombay.

Projects

Chakuli — Offline AI Chatbot

Aug 2025 – Present

Kotlin, Jetpack Compose, LiteRT-LM, Hilt, CameraX, Proto DataStore

- **Closed beta on Google Play.** On-device Android AI assistant running Gemma 2/3, Phi 3.5 Mini, SmolLM entirely via **LiteRT-LM** — no internet for inference, complete data privacy.
- Agent Skills system, audio chat, in-app camera (CameraX), foreground WorkManager for large model downloads, Protobuf DataStore for conversation persistence, RevenueCat tip jar.

OCanPredict — OSMF Screening App

Dec 2022 – Jul 2023

Kotlin, Java, MediaPipe FaceMesh, Firebase

- Android diagnostic app using **MediaPipe FaceMesh** for real-time OSMF facial screening — increased diagnostic efficiency by **40%**. Won multiple national and state-level awards; contributed to granted patent.

V-Safe-Anywhere: Women's Safety Wearable

Nov 2022 – Apr 2024

Flutter, Firebase, IoT, Google Cloud

- Flutter app paired with a custom IoT wearable: panic button triggers instant **live GPS broadcast** to emergency contacts via Firebase — **98% uptime**. Published by [IGI Global](#).

Autonomous Garbage Collection Robot

Jun 2023 – Sep 2023

Python, YOLOv5, OpenCV

- Computer vision system (YOLOv5) for autonomous floating waste detection — reduced manual effort by **60%**. Published at [8th ICACS 2024](#), ACM, Hong Kong.

Publications, Patents & Conferences

Journal: “V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT.” *IGI Global*, Mar 2024. [\[Link\]](#)

Conference: “Autonomous Floating Garbage Collection Device using Computer Vision and Robotics.” *8th ICACS 2024*, ACM, Hong Kong, Oct 2024. [\[Link\]](#)

Conference: “Development of a Mobile Field Trip Application.” *3rd Annual TRESL Lab International Symposium*, Athabasca, Canada, Jul 2024.

Patent (Granted): “Smart Chairside TMJ Examination and Muscle Activity Detector Tool.” Office of the Controller General of Patents, India. (Pub. No. 202321022476, Filed Mar 2023 — **Granted**.)

Patent (Pending): “An Artificial Intelligence Based Portable Device for Dental Disease Detection.” Govt. of India — DPIIT. (App. No. 202421006731, Filed Feb 2024, Published Aug 2025 — Awaiting Examination.)

Awards & Recognition

3rd Place — Avishkar 2024 **Runner-up** — DIPEX 2023 **Runner-up** — SHRISTI 2023

Grant Awardee — BIREC E-YUVA Grant, Rs. 10 lakh seed funding (Feb 2023)

Winner — MEDHA 2022 **Winner** — MEDIC-2022 Multiple national awards for OCanPredict.

Writing & Open Source

Medium (medium.com/@bhanu.r.nagpure.8082): Technical blogger on indie Android/Flutter dev — production postmortems and engineering decisions.

Open Source: Contributor to **Wikimedia Foundation / Scribe-Android** (Kotlin) — tablet keyboard layouts for 7+ languages, UI bug fixes, KeyHandler refactor.